

Repetition

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```
print(1)
print(2)
print(3)
print(4)
print(5)
print(6)
```

```
for i in range(1, 100):
    print(i)
```

start
stop-1

This kind of repetition is also called iteration.
Repetition of a set of statements in a program is made possible using looping constructs.

Syntax of the for loop

```
for <control_variable> in range(start, stop):
    <statements inside body of the for loop>
```

For example:

```
n = int(input('Enter number of terms: '))
for i in range(1, n):
    print(i, end=' ')
```

Output:

```
Enter number of terms: 31
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
```

Syntax of the for loop with skip value or jump value

```
for <control_variable> in range(start, stop, skip_value(default value is
1, if this is omitted):
    <statements inside body of the for loop>
```

For example:

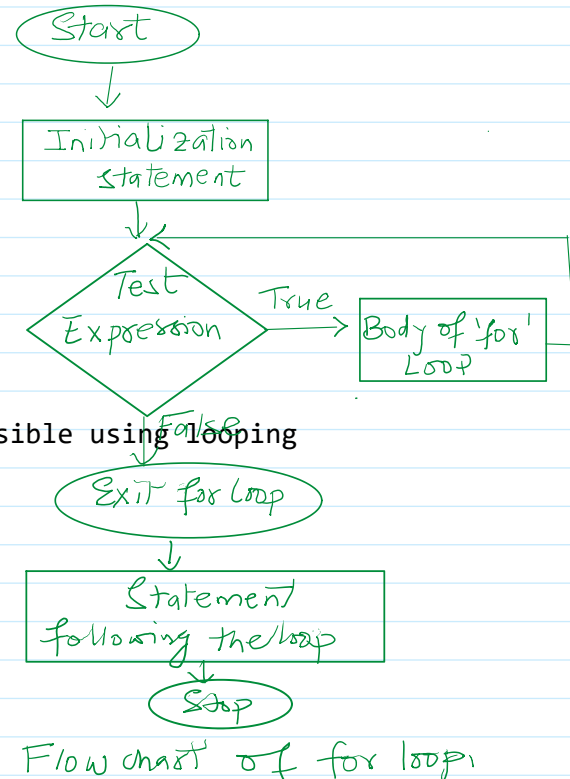
```
n = int(input('Enter number of terms: '))
for i in range(1, n, 2):
    print(i, end=' ')
```

Output:

```
Enter number of terms: 31
1 3 5 7 9 11 13 15 17 19 21 23 25 27 29
```

Another example:

```
start = int(input('Enter start value: '))
end = int(input('Enter end value: '))
sv = int(input('Enter skip or Jump: '))
```



```
for i in range(start, end+1, sv):
    print(i, end=' ')
```

Output:

```
Enter start value: 10
Enter end value: 100
Enter skip or Jump: 5
10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
```

Reverse:

```
start = int(input('Enter start value: '))
end = int(input('Enter end value: '))
sv = int(input('Enter skip or Jump: '))
for i in range(start, end+1, sv):
    print(i, end=' ')

print('\nReverse: ')
for i in range(end, start-1, -sv):
    print(i, end = ' ')
```

Output:

```
Enter start value: 10
Enter end value: 100
Enter skip or Jump: 5
10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Reverse:
100 95 90 85 80 75 70 65 60 55 50 45 40 35 30 25 20 15 10
```

WAP to print n terms of odd numbers divisible by 3.

Input: Enter number of terms: 5

Output: 3, 9, 15, 21, 27

Homework:

1. Write a Python program to print numbers from 1 to 50 using a for loop.
2. Write a Python program to print the first n natural numbers using:
for i in range(1, n+1)
3. Write a Python program to print all even numbers between 1 and 100 using a for loop.
4. Write a Python program to print all odd numbers between 1 and 50 using:
range(start, stop, skip_value)
5. Write a Python program to print multiplication table of a given number using a for loop.

Example:

Input: 5

Output:

5 x 1 = 5

5 x 2 = 10

...

5 x 10 = 50

6. Write a Python program to print numbers from 100 to 1 in reverse order using:
range(start, stop, -1)
7. Write a Python program to print numbers from 10 to 100 with a jump value of 10.

Expected Output:

10 20 30 40 50 60 70 80 90 100

8. Write a Python program to print the first n numbers divisible by 5.
Example:
Input: 6
Output:
5 10 15 20 25 30
9. Write a Python program to print all numbers between two user-given values using a user-defined skip/jump value.
Example:
Input:
Start = 2
End = 20
Jump = 3
Output:
2 5 8 11 14 17 20
10. Write a Python program to print the reverse of a sequence entered by the user using a negative skip value.
Example:
Input:
Start = 1
End = 20
Jump = 2
Output:
19 17 15 13 11 9 7 5 3 1
11. Write a Python program to print the first n multiples of 7 using a for loop.
Example:
Input: 5
Output:
7 14 21 28 35
12. Write a Python program to print the first n odd numbers divisible by 3.
Example:
Input: 5
Output:
3 9 15 21 27